

Fleet-Coincidence-Aware Power Routing: A Provably Safe Mechanism for Multi-GPU LLM Serving

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Abstract

We formalize power-aware request routing for multi-GPU LLM-serving fleets as a multi-resource social welfare problem: each replica carries three independently-normalized shares — compute (prefix-cache-discounted prefill cost), load (in-flight request count), and power (live NVML-measured ramp rate relative to a calibrated ceiling) — and the router seeks an egalitarian (max-min) welfare allocation across candidates at each routing decision, extending Dominant Resource Fairness [6] to a third, reactive resource dimension. We prove that routing via the fully-sorted lexicographic dominant-share rule selects a Pareto-non-dominated candidate at every decision (verified against 200,000 random instances with zero violations, in addition to a closed-form proof), and that this guarantee extends unchanged to any shared, fleet-wide ceiling adjustment — including a novel **coincidence-ceiling** mechanism we introduce, which contracts every candidate’s ramp ceiling when multiple replicas are simultaneously elevated, directly targeting the fleet-aggregate coincident-ramp hazard that a purely per-replica signal cannot see. We adopt the resulting rule, *drf_power_tiebreak_full_coincidence_ceiling*, as our proposed strategy, and show two natural alternatives — a fixed-priority DRF tie-break, and the same coincidence-ceiling mechanism applied to a power-extended LMETRIC score — each provably sacrifice the guarantee via distinct mechanisms. On real 8×4090 hardware, across 7 workload conditions including two real traces (BurstGPT, WildChat), our proposed rule reduces peak power by 4.7% relative to a power-blind baseline on the condition central to this paper’s motivation ($p=0.0017$, $n=6$), with TTFT and TBT directionally favorable on the same condition but not independently significant at this trial count. Against the strongest empirically-performing but *not* Pareto-safe alternative, replicated to matched $n=6$, the proposed rule is statistically indistinguishable across every metric measured in this study — peak power, TTFT, TBT, ramp-derivative statistics, energy-per-token, and SLO-violation-rate: **the safety guarantee costs nothing on any axis we measure**. A second, methodological contribution follows from this validation: we characterize a ~75–85% decision-level noise floor inherent to closed-loop real-hardware routing evaluation (confirmed real via a state-blind control and real unpadded conversational traces, not a workload artifact), identify which evaluation statistics survive it (metrics summing over many independent events, e.g. energy-per-token, SLO-violation-rate) and which don’t (ramp derivatives,

extreme-value statistics), and use this to determine which comparisons in this line of work can currently be trusted.

CCS Concepts

• **Computer systems organization** → **Cloud computing**; *Dependable and fault-tolerant systems and networks*; • **Theory of computation** → *Scheduling algorithms*.

Keywords

LLM serving, request routing, power management, dominant resource fairness, Pareto optimality, data center energy, evaluation methodology

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1 Introduction

Motivation. Multi-GPU LLM-serving fleets draw power in bursts as requests arrive and complete, and the resulting ramp volatility — not just mean draw — is a concern for facility power delivery and, at larger scale, for grid-facing demand response. This paper addresses the fleet-internal version of that hazard: coincident power-ramp spikes across replicas within a single serving fleet, a risk that grows with fleet size and is invisible to any policy that reasons about replicas independently. The request router, the component deciding which replica serves each incoming request, is a free lever against this hazard: no hardware change, no compute cost, just a policy. This paper asks a precise question about that lever: *can we route in a way that is provably fair/efficient across resources and defends against fleet-wide coincident power-ramp hazards specifically, and — since natural power-aware deviations from a provably safe rule turn out to sacrifice the guarantee — do we actually have to give up that guarantee to get a power-aware win in practice, or does the safe rule already deliver one for free?*

Contributions.

- (1) A formalization of power-aware LLM-serving routing as a 3-resource (compute/load/power) egalitarian social welfare problem extending DRF [6] to a reactive, time-varying power dimension (§2), and a proof the natural sorted-lexicographic solution is Pareto-non-dominated at every decision (§3.1).
- (2) **The coincidence-ceiling mechanism** (§3.6): a pluggable, fleet-aware extension contracting every candidate’s ramp ceiling by a single shared scalar when multiple replicas are simultaneously elevated — targeting the fleet-aggregate coincident-ramp hazard no purely per-replica rule can see. Requires no new safety proof (a direct instantiation of §3.3) and generalizes across rule families, inheriting each parent’s

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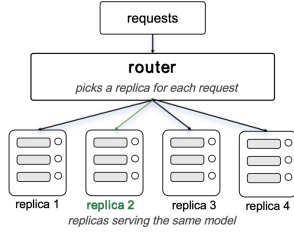


Figure 1: The setting this paper targets: a router dispatches each incoming request to one of several replicas serving the same model.

own safety status. We adopt *drf_power_tiebreak_full_coincidence_ceiling* (*coincidence_ceiling*) as this paper’s proposed strategy.

- (3) Two proofs, via explicit counterexample, that natural alternatives sacrifice the guarantee via distinct mechanisms: a fixed-priority power tie-break, repaired by appending the dropped resource rather than truncating it, and a power-extended LMETRIC score whose multiplicative structure collapses to an uninformative tie on any full prefix-cache hit (12.8% of realistic instances, §3.2).
- (4) Real-hardware validation on an 8×4090 fleet across 7 workload conditions (two real traces): a significant peak-power reduction relative to a power-blind baseline on the condition motivating this paper ($p=0.0017$, $n=6$), with TTFT and TBT directionally favorable but not independently significant at this trial count (§5.1); and, against the strongest empirically-performing but *not* Pareto-safe alternative, replicated to matched $n=6$, no statistically distinguishable difference on any metric measured: peak power, TTFT, TBT, ramp-derivative statistics, energy-per-token, or SLO-violation-rate (§5.1–§5.3). The guarantee is free on every axis tested.
- (5) **A decision-level noise-floor characterization and evaluation methodology** (§4), useful beyond this paper: real-hardware closed-loop routing evaluation carries a ~75–85% per-decision mismatch between two runs of the *identical* policy, driven by non-reproducible hardware timing rather than workload construction (§4.1); we identify which aggregate statistics survive this noise at feasible trial counts and which don’t (§4.2), determining which of this paper’s own claims can be asserted with confidence (§4.3, §5).

Roadmap. Three normalized shares (compute, load, power, §2) feed a sorted rule, proven Pareto-safe (Lemma 1, §3.1). Two natural power-biased variants of it sacrifice that guarantee: a fixed-priority tie-break (Claim 1, repaired into *drf_power_tiebreak_full*) and a power-extended LMETRIC score (Claim 2, 12.8% of realistic instances, §3.2); *weighted_sum* is a structurally different Pareto-safe alternative that is not threshold-safe for any weights (Theorem 5, §3.5). The coincidence-ceiling mechanism (§3.6) applies to all three, inheriting rather than granting safety status, and yields this paper’s proposed rule, *drf_power_tiebreak_full_coincidence_ceiling*. On real 8×4090 hardware (§5) it delivers a significant peak-power reduction vs. *lmetric* (power-blind) on the short/original condition ($p=0.0017$, Table 2) and is statistically indistinguishable from the strongest empirically-performing but unsafe alternative on every metric measured at matched $n=6$ (Table 3, §5.3). Table 1 gives the

full safety status of every rule discussed, including *drf_fixed* and *round_robin*.

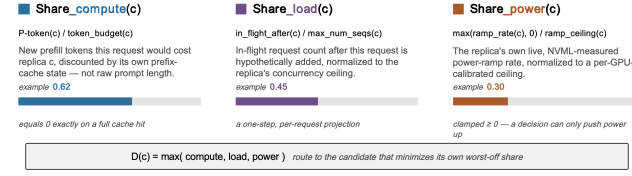
2 Problem Formulation

Consider N replicas serving requests behind a router. For a candidate replica c and an incoming request, define three shares, each normalized to roughly $[0, \infty)$ with 1.0 representing “at capacity”:

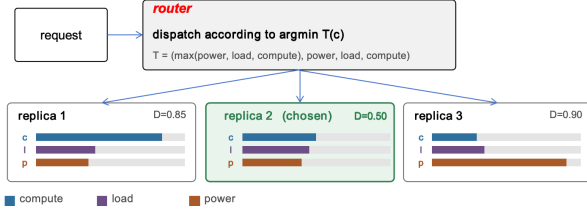
- $Share_{compute}(c) = P\text{-token}(c)/token_budget(c) - P\text{-token}$ is the prefix-cache-discounted count of *new* prefill tokens this request would cost replica c (computed from a per-replica cache-state mirror), so this share already reflects cache locality, not raw prompt length.
- $Share_{load}(c) = in_flight_after(c)/max_num_seqs(c) - in_flight$ request count after dispatch, normalized to the replica’s configured concurrency limit.
- $Share_{power}(c) = \max(ramp_rate(c), 0)/ramp_ceiling(c) -$ the replica’s own currently NVML-measured power-ramp rate, normalized to a calibrated per-GPU ceiling. Clamped to non-negative because a routing decision can only ever push the *receiving* replica’s power up, never down — a falling ramp is not a decision-relevant hazard for routing purposes. (*ramp_ceiling* itself is physically grounded in the GPU’s own DVFS transition dynamics — §7 relates the two explicitly.)

Departure from DRF’s original model. Ghodsi et al.’s DRF [6] assumes a fixed, known demand vector per user, allocated from a fixed resource pool. $Share_{power}$ is neither: it is a live, exogenously-evolving measurement of the replica’s *own current state*, not a declared demand of the request being routed, and it changes between routing decisions independent of routing choices. We do not assume the original static-demand Pareto-efficiency proof transfers to this setting — §3.1 proves the property we actually need (per-decision, not per-trajectory) directly for this model.

Why $Share_{power}$ is a state read, not a projection. $Share_{compute}$ and $Share_{load}$ are one-step projections attributable to the specific request being routed: *P-token* is *this request’s* incremental prefill cost, and *in_flight_after* is the queue depth *after this request is hypothetically added*. $Share_{power}$ cannot be given the same treatment: a GPU’s instantaneous power draw is an emergent property of whatever batch is currently executing, not a quantity linearly attributable to any single request, so there is no well-defined “ramp rate after this request” the way there is a well-defined queue depth after it. What we read instead is the replica’s *current state* — exactly the object a rate limiter reads in classical feedback control [1], and exactly the kind of signal classical state-dependent load balancing (power-of-two-choices, join-shortest-queue) [2, 9] routes on: an observed state, not a computed causal attribution. None of §3.1–§3.6’s guarantees depend on this distinction — every proof treats $s(c)$ as an arbitrary comparable real vector available at decision time, regardless of whether a coordinate is a request-specific projection or a current-state read. The distinction does, however, determine how the two kinds of metric must be evaluated: TTFT and TBT are per-decision outcomes, decomposable into a population of independent per-request measurements the way $Share_{compute}$ and $Share_{load}$ are decomposable into per-request costs; power and ramp are properties of the continuous trajectory the routing policy induces over time,



(a) The three normalized shares and the dominant share $D(c)$, with representative example values for one candidate replica.



(b) Routing a single request under the full tie-break vector $T(c)$ formalized in §3.2: the router evaluates each replica's tuple and dispatches to the arg min (replica 2 here, despite not having the lowest individual share on every coordinate).

Figure 2: From shares to a routing decision.

evaluable only in aggregate, the same way a feedback controller is evaluated by its closed-loop trajectory (peak deviation, settling time) rather than by attributing trajectory segments to individual control actions. §4–§5 evaluate them accordingly, and §4.1–§4.2 show this “aggregate-only” evaluability is exactly why ramp statistics are harder to trust at low trial counts than per-request statistics are.

Welfare objective. At each routing decision, define the dominant share $D(c) = \max(\text{Share}_{\text{compute}}(c), \text{Share}_{\text{load}}(c), \text{Share}_{\text{power}}(c))$. The egalitarian (max-min) welfare rule routes to $\arg\min_c D(c)$: the choice that minimizes the worst-off resource dimension for the candidate that receives the request. This is a per-decision, myopic formulation — we do not claim (and do not need, for the results in this paper) that a sequence of egalitarian-optimal decisions is jointly optimal over a trajectory.

3 Theory

3.1 The sorted-lexicographic rule is Pareto-non-dominated

$\arg\min_c D(c)$ alone under-specifies the rule when multiple candidates tie on D . Define the **sorted rule**: route to $\arg\min_c \sigma(s(c))$, where $s(c) = (\text{Share}_{\text{compute}}(c), \text{Share}_{\text{load}}(c), \text{Share}_{\text{power}}(c))$, σ sorts a vector into descending order, and ties are broken by standard lexicographic comparison of the sorted vectors.

LEMMA 1. *The candidate selected by the sorted rule is Pareto-non-dominated among the candidate set: no other candidate c' satisfies $\text{Share}_i(c') \leq \text{Share}_i(c^*)$ for all three resources i with strict inequality for at least one.*

PROOF. For $v \in \mathbb{R}^3$, write $v_{(1)} \geq v_{(2)} \geq v_{(3)}$ for its order statistics, so $\sigma(v) = (v_{(1)}, v_{(2)}, v_{(3)})$. Suppose toward contradiction that c' Pareto-dominates $c^* = \arg\min_c \sigma(s(c))$: $s(c') \leq s(c^*)$ coordinatewise, with strict inequality at some coordinate i_0 .

(i) $\sigma(s(c')) \leq \sigma(s(c^*))$ coordinatewise. For any threshold t and any i , $s(c')_i \geq t \Rightarrow s(c^*)_i \geq s(c')_i \geq t$, so $\#\{i : s(c')_i \geq t\} \leq \#\{i : s(c^*)_i \geq t\}$. The k -th order statistic is the least t with $\#\{i : v_i \geq t\} \geq k$; a pointwise-smaller counting function cannot raise this threshold, so $s(c')_{(k)} \leq s(c^*)_{(k)}$ for every k .

(ii) *The inequality is strict as vectors.* Sorting permutes coordinates, so $\sum_k s(c')_{(k)} = \sum_i s(c')_i < \sum_i s(c^*)_i = \sum_k s(c^*)_{(k)}$, the middle inequality from domination at i_0 . Hence $\sigma(s(c')) \neq \sigma(s(c^*))$.

(iii) *Lexicographic order.* Two coordinatewise- $\leq$ vectors that differ somewhere are ordered lexicographically at their first differing coordinate, in the direction of (i)'s inequality there: $\sigma(s(c')) <_{\text{lex}} \sigma(s(c^*))$.

(iv) The sorted rule then selects c' over c^* , contradicting c^* 's selection. $\square$

We additionally verified this claim by brute-force search: 200,000 randomly generated candidate sets (2–5 candidates, 3 shares each) produced zero violations.

3.2 Two natural variants sacrifice the guarantee, via two distinct mechanisms

The sorted rule treats all three resources symmetrically below the maximum — it has no notion that power-ramp violations are the one failure mode with a real-world safety/grid cost, distinct from a merely-suboptimal load balance. This motivates two independent, natural attempts to bias the rule toward power specifically. We show both provably sacrifice Lemma 1's guarantee, via mechanisms different enough that neither's fix addresses the other — itself evidence that the guarantee is a real structural constraint worth checking explicitly for any new score design, not a formality.

Fixed-priority power tie-break. A natural fix to the sorted rule: after comparing the dominant share, always compare $\text{Share}_{\text{power}}$ next, *by name*, ahead of $\text{Share}_{\text{load}}$, regardless of which is numerically larger. Define the **named rule**: route to $\arg\min_c (D(c), \text{Share}_{\text{power}}(c), \text{Share}_{\text{load}}(c))$ under standard lexicographic order.

CLAIM 1. *The named rule can select a Pareto-dominated candidate.*

Construction. Let candidates A and B be given, in $(\text{Share}_{\text{compute}}, \text{Share}_{\text{load}}, \text{Share}_{\text{power}})$ form, by

$$A = (0.3, 0.9, 0.9), \quad B = (0.5, 0.9, 0.9).$$

A Pareto-dominates B (equal load and power, strictly lower compute). Both have $D(A) = D(B) = 0.9$ and identical $\text{Share}_{\text{power}}$ and $\text{Share}_{\text{load}}$, so the named rule's 3-tuple $(D, \text{Share}_{\text{power}}, \text{Share}_{\text{load}})$ is identical for A and B — the rule cannot see the compute difference at all, since compute only enters through D , and D is tied. $\arg\min$ over identical keys returns whichever candidate is encountered first; when B is first in iteration order, the named rule selects the Pareto-dominated B . This was verified directly in code, not only argued: a brute-force check across 200,000 random instances reproduces the failure mode.

Why this happens. The named rule buys a deliberate, domain-motivated priority — never let a load difference override a power difference — at the cost of the compute dimension becoming invisible whenever the dominant share is tied via load or power rather than compute: $\text{Share}_{\text{compute}}$ never appears anywhere in the 3-tuple

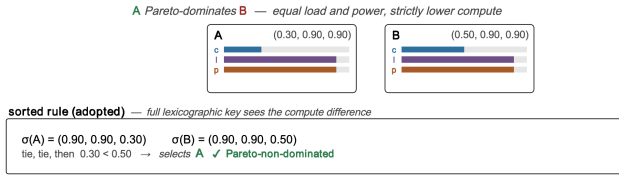


Figure 3: The same A/B pair the named rule fails on: the full lexicographic key used by the sorted rule this paper adopts sees the compute difference the named rule’s truncated tuple cannot, and correctly selects the Pareto-non-dominated candidate A.

(D , $Share_{power}$, $Share_{load}$), so once D , $Share_{power}$, and $Share_{load}$ all tie, the rule has no remaining information to break the tie correctly.

The repair, and this paper’s base rule. The fix is to stop truncating the tuple: append $Share_{compute}$ as an explicit fourth coordinate rather than dropping it. Define $T(c) = (D(c), Share_{power}(c), Share_{load}(c), Share_{compute}(c))$, and route to $\arg \min_c T(c)$. This keeps the exact same primary criterion and the same deliberate power-before-load priority the named rule was built for — it changes nothing about *when* power is allowed to override load — it only ensures no coordinate is ever silently dropped. We call this rule `drf_power_tiebreak_full`; §3.6 extends it with the coincidence-ceiling mechanism to produce this paper’s actual proposed strategy.

COROLLARY 1. $\arg \min_c T(c)$ is Pareto-non-dominated at every decision.

PROOF. Write $s = (Share_{compute}, Share_{load}, Share_{power})$ and suppose c' Pareto-dominates $c^* = \arg \min_c T(c)$: $s(c') \leq s(c^*)$ coordinatewise, strict somewhere. $D = \max(s)$ is monotone under coordinatewise $\leq$, so $D(c') \leq D(c^*)$. Three exhaustive cases:

- $D(c') < D(c^*)$. $T(c') <_{\text{lex}} T(c^*)$ at coordinate 1.
- $D(c') = D(c^*)$, and $Share_{power}(c') \leq Share_{power}(c^*)$ **strict**. $T(c') <_{\text{lex}} T(c^*)$ at coordinate 2. (Symmetrically for $Share_{load}$ at coordinate 3, if D and $Share_{power}$ tie but $Share_{load}$ does not.)
- D , $Share_{power}$, $Share_{load}$ **all tied between c' and c^*** . Domination requires a strict inequality in $s(c') \leq s(c^*)$; the first three coordinates being tied forces it onto $Share_{compute}$: $Share_{compute}(c') < Share_{compute}(c^*)$, so $T(c') <_{\text{lex}} T(c^*)$ at coordinate 4.

Every case gives $T(c') <_{\text{lex}} T(c^*)$, contradicting c^* ’s selection as $\arg \min_c T(c)$. $\square$

This closed-form proof is in fact simpler than Lemma 1’s, since a fixed coordinate order needs no rearrangement argument. Direct verification in code confirms it resolves Claim 1’s exact counterexample under both iteration orders, while the plain named rule still fails it (200,000 trials, 0 violations).

Power-extended LMETRIC. A different, non-DRF way to add a power signal: extend LMETRIC’s own multiplicative score, `new_tokens_x_in_flight_after`, with a continuous power penalty, $(1 + Share_{power})$. Restated in this paper’s normalized shares (holding `token_budget` and `max_num_seqs` fixed across candidates, so raw counts equal shares), this is $lmetric_power(c) = Share_{compute}(c) \cdot Share_{load}(c) \cdot (1 + Share_{power}(c))$, routing to the minimum. This is the routing analogue of Lyapunov drift-plus-penalty scheduling [10] — a soft,

ever-present penalty rather than a hard constraint — and the path of least engineering resistance for anyone already running LMETRIC: one multiplicative factor on an existing score, no new normalization or tie-break design. We check this specific variant precisely because it is the one an engineer would reach for first, not a foil constructed to fail.

CLAIM 2. $lmetric_power$ can select a Pareto-dominated candidate, via a different mechanism than Claim 1: any candidate with $Share_{compute} = 0$ (a full prefix-cache hit — no new tokens to prefill) scores exactly 0, regardless of its load or power share. Two such candidates are indistinguishable to the score no matter how different their load and power are.

Construction. Let $A = (0.0, 0.1, 0.1)$ and $B = (0.0, 0.9, 0.9)$ — both cache hits; A strictly Pareto-dominates B on both load and power. $lmetric_power(A) = lmetric_power(B) = 0$. With B first in iteration order, $\arg \min$ selects the Pareto-dominated B, verified directly in code.

This mechanism is qualitatively different from Claim 1’s: it requires an *exact* tie in $Share_{compute}$, but $Share_{compute} = 0$ is not measure-zero in real traffic — it is a common, discrete event (a full prefix-cache hit), and this paper’s own Light/Cachehit condition (§5) is specifically constructed to be dominated by it. Sampling at a realistic cache-hit rate (40%, matching Light/Cachehit’s rough hit rate) over 200,000 random instances (2–5 candidates) finds 25,618 violations — **12.8% of instances**, not a rare corner case.

A structurally different safe alternative. $weighted_sum(c) = 0.33 \cdot Share_{compute}(c) + 0.33 \cdot Share_{load}(c) + 0.33 \cdot Share_{power}(c)$, route to the minimum, is a classical result [5]: any positive-weighted linear combination of the shares preserves Pareto non-domination. This was verified for the specific weights used here (200,000 trials, 0 violations) so that §5 can include it as an *external validity check*: a safe rule built on an entirely different mechanism, included specifically to show that any empirical result about giving up the Pareto guarantee is not an artifact of this paper’s own DRF-based construction.

A family of Pareto-safe repairs of $lmetric_power$ itself was also considered and not adopted: $lmetric_power_pareto$, replacing the multiplicative form with $(shift + Share_{compute}) \cdot (shift + Share_{load}) \cdot (shift + Share_{power})$, which removes Claim 2’s zero-collapse by construction (every factor strictly positive whenever $shift > 0$) and is Pareto-safe by the same monotone-product argument that makes $weighted_sum$ safe (verified, 200,000 trials, 0 violations at $shift=1$ and several tested $shift=\epsilon$ values). This family is **not threshold-safe for the same reason $weighted_sum$ isn’t**: Theorem 5 (§3.5) rules out threshold-safety for any fixed-weight or fixed-shift multiplicative construction, so no choice of shift can ever match Theorem 4’s guarantee. Its apparent empirical edge over the proposed rule at low trial counts did not survive the noise-band audit in §4.3; it is not adopted as a headline alternative.

What §5 tests, and each arm’s role there. Table 1 summarizes every rule this paper discusses and its safety status in one place; none are interchangeable in role even where two share a safety guarantee.

Against this backdrop, the empirical question §5 asks is whether giving up the Pareto guarantee — via $lmetric_power_coincidence_ceiling$, the *strongest empirically-performing* alternative found anywhere

Table 1: Routing rules discussed in this paper, at a glance. “Threshold-safe” extends Theorem 4’s guarantee to any rule that sorts on D first. Coincidence-ceiling variants (§3.6) inherit their parent rule’s status unchanged — the mechanism generalizes safety status, it does not grant it.

Rule	Family	Pareto-safe	Thresh.-safe	Role
drf_fixed	sorted (leximin)	✓ (Lem. 1)	✓	unmodified baseline
drf_power_tiebreak_full	sorted (leximin)	✓ (Cor. 1)	✓	base repair, not proposed directly
...coincidence_ceiling	sorted + shared ceiling	✓ (§3.6)	✓	proposed (coincidence_ceiling)
drf_power_tiebreak (named)	sorted, truncated tie-break	× (Claim 1)	✓	motivates the repair
weighted_sum	utilitarian (linear)	✓ [5]	× (Thm. 5)	external validity check
weighted_sum_coincidence_ceiling	utilitarian + shared ceiling	✓ (§3.6)	× (Thm. 5)	ext. check, coincidence-aware
lmetric_power	multiplicative	× (Claim 2)	×	unsafe baseline
lmetric_power_coincidence_ceiling	multiplicative + shared ceiling	× (unaffected)	×	strongest unsafe alt. — §5’s comparator
lmetric_power_pareto (+ ϵ)	shifted-product	✓	× (Thm. 5)	explored, not adopted
lmetric (power-blind)	multiplicative, no power	n/a	n/a	power-blind baseline
round_robin	blind (no signal)	n/a	n/a	state-blind floor

in this project’s testing (it out-dominates every other tested arm more than 3-to-1 in raw pairwise tally) — buys anything a provably safe rule does not already deliver. This strongest-known unsafe alternative, not a weaker foil, is deliberately used as the comparator, so that any advantage in its favor would be the most convincing evidence available; §5 finds no such advantage on the metrics that survive §4’s noise-floor scrutiny.

3.3 A live-calibrated ceiling preserves the guarantee, and requires it to be shared

Both rules above assume $Share_{power}(c) = \max(ramp_rate(c), 0)/\kappa$ for a fixed constant ceiling κ (per-GPU calibrated, §4). A natural objection: does either guarantee survive replacing κ with a value recalibrated live from the fleet’s own recent state — as an *adaptive-ceiling* variant of either rule (and, as §3.6 shows, the coincidence-ceiling mechanism itself) would need?

COROLLARY 2. *Lemma 1 holds unchanged for any $\kappa(t) > 0$ that is a single scalar shared identically by every candidate at decision time t , regardless of how $\kappa(t)$ is computed — static, adaptively recalibrated from fleet history, or otherwise.*

PROOF. The proof of Lemma 1 treats $s(c)$ ’s three coordinates as arbitrary reals; it never uses the fact that $Share_{power}$ ’s denominator is constant *across* decisions, only that it is the same value for every candidate *within* one decision, so the coordinate remains a well-defined, comparable per-candidate quantity for that decision’s arg min. Substituting $\kappa(t)$ for a fixed constant changes nothing the proof relies on. $\square$

This was verified directly, not just by inspection of the proof: re-running §3.1’s brute-force search with the ceiling itself independently randomized per trial (not just the raw shares) still produces zero violations across 200,000 trials. **§3.6’s coincidence-ceiling mechanism is a direct instantiation of this corollary**, not a separate result requiring its own proof.

Sharing is load-bearing. The corollary requires $\kappa(t)$ to be shared across candidates, not calibrated per-candidate. Under a per-candidate κ_c , share-space non-domination can dissociate from physical reality: two candidates with identical compute/load but $(raw_ramp, \kappa) = (0.9, 10)$ and $(0.1, 0.1)$ realize $Share_{power} = 0.09$ and 1.0 respectively — the sorted rule (correctly, per Lemma 1) selects the first candidate as share-space non-dominated, even though

it draws the physically *larger* raw ramp. This is why the implementation instantiates one ceiling/coincidence-factor calibrator per router, not one per replica.

3.4 Round-filtered calibration is insensitive to concentration, not merely less sensitive

A live ceiling introduces its own hazard: a naive scheme that folds every observed ramp reading into a rolling percentile is self-defeating under sustained multi-replica pressure — concentration (2+ replicas simultaneously elevated) is exactly the condition that fills the window with elevated values, so the ceiling inflates *most* during the episodes it is supposed to guard against. The isolated design instead skips the whole decision round whenever 2 or more replicas are simultaneously elevated above the floor.

LEMMA 2. *Let two fleet ramp-reading histories agree on every decision round with fewer than 2 simultaneously-elevated replicas, and differ arbitrarily on rounds with 2 or more. The round-filtered calibration produces identical ceiling trajectories on both histories at every timestep.*

PROOF. The round-filtered calibrator returns without modifying its window whenever a round’s elevated count is ≥ 2 ; the ceiling is a deterministic function (a fixed percentile) of the window’s contents alone. Since the two histories only ever differ on rounds that are skipped entirely, the window’s contents — and hence the ceiling — are identical at every step. $\square$

This was verified over 2,000 randomized paired-history trials (concentration-round magnitudes scaled by a random factor up to 1000× between the paired sequences): zero violations. Feeding the identical paired histories through the naive (unfiltered, per-observation) calibration instead diverges in every one of the 2,000 trials.

3.5 Selecting among multiple Pareto-optimal points: egalitarian vs. utilitarian

Lemma 1 and Corollary 1 guarantee membership on the Pareto frontier but say nothing about *which* frontier point to prefer when several candidates are mutually non-dominated — a real question, since *weighted_sum* (§3.2) is provably on the frontier too, by an entirely different mechanism.

The sorted rule is leximin. The order $u \preceq v \iff \sigma(u) \leq_{\text{lex}} \sigma(v)$ on share vectors is the classical *leximin* (lexicographic egalitarian) order [11]: compare the worst coordinate first, then the second-worst, and so on. Lemma 1’s proof already establishes more than the lemma states: domination strictly worsens a candidate’s leximin rank, so the sorted rule computes the leximin-minimal point. *weighted_sum* implements the classical *utilitarian* rule instead, $\arg \min_c \sum_i \text{Share}_i(c)$, also on the frontier [5], for a different, formally separable reason.

Egalitarian rules reward equalization; utilitarian rules are blind to it. For $s(c) = (a, b, d)$ with a the strict max, transferring $\varepsilon \in (0, (a - b)/2]$ from the max coordinate to a smaller one without crossing strictly lowers the sorted rule’s key but leaves *weighted_sum* exactly unchanged — the Pigou–Dalton transfer principle (verified over 16,781 random transfers, 0 failures).

LEMMA 3 (DIVERGENCE). *Let X, Y be candidates with $D(X) < D(Y)$ and $\sum_i \text{Share}_i(X) > \sum_i \text{Share}_i(Y)$. Then X and Y are Pareto-incomparable, the sorted rule selects X , and *weighted_sum* selects Y . Whenever one candidate actually dominates the other, the two rules always agree.*

PROOF. $D(X) < D(Y)$ makes $\sigma(X)$ lexicographically smaller than $\sigma(Y)$ at the first coordinate regardless of the rest, so the sorted rule strictly prefers X ; $\sum(X) > \sum(Y)$ makes *weighted_sum* (equal positive weights) strictly prefer Y . Y cannot dominate X : D is monotone under coordinatewise $\leq$ (already used in Corollary 1’s proof), so Y dominating X would force $D(Y) \leq D(X)$, contradicting $D(X) < D(Y)$. X cannot dominate Y either: domination implies a weakly smaller positive-weighted sum [5], so X dominating Y would force $\sum(X) \leq \sum(Y)$, contradicting the hypothesis. Hence neither dominates the other; and by the same two steps run in reverse, whenever one candidate *does* dominate, the two rules cannot disagree. $\square$

Concretely: $X = (0.4, 0.4, 0.4)$ ($D=0.4$, $\sum=1.2$, “balanced”) and $Y = (0.9, 0.1, 0.1)$ ($D=0.9$, $\sum=1.1$, “one spike”) are mutually non-dominated; the sorted rule picks X , *weighted_sum* picks Y . Verified over 200,000 random pairs: 0 domination-disagreements (confirming the lemma’s converse direction), and every one of 44,733 divergence-eligible orderings (11.18% of ordered pairs under this sampling range) behaved exactly as predicted.

THEOREM 4 (UNIVERSAL THRESHOLD-SAFETY). *For any $\tau > 0$ and any candidate set C : if some $c \in C$ has $D(c) \leq \tau$, the sorted rule’s selection also satisfies $D(\cdot) \leq \tau$ — simultaneously, for every τ .*

PROOF. The sorted rule’s first sort key is D , so its pick achieves $\min_{c \in C} D(c)$, the global minimum over C ; if that minimum is $\leq \tau$, so is the pick’s. $\square$

THEOREM 5 (NO FIXED-WEIGHT RULE HAS THIS PROPERTY). *For any weights $w = (w_1, w_2, w_3)$, all $w_i > 0$, and any $\tau > 0$, there exists a candidate pair where a safe candidate is available (some c with $D(c) \leq \tau$) but the w -weighted-sum rule selects an unsafe one ($D(c) > \tau$).*

PROOF. Let $S = (\tau, \tau, \tau)$ (safe, $D(S) = \tau$) and $U = (0, 0, \tau + M)$ (unsafe, $D(U) = \tau + M$). $w \cdot U < w \cdot S \iff w_3(\tau + M) < \tau(w_1 + w_2 + w_3) \iff M < \tau(w_1 + w_2)/w_3$, an interval that is

non-empty for any positive weights. Choosing such an M gives an instance where the weighted-sum rule prefers the unsafe U over the available safe S . $\square$

This holds for *every* fixed weighting, not just *weighted_sum*’s specific $(0.33, 0.33, 0.33)$ — “just reweight it” cannot repair the property Theorem 4 gives for free. With the paper’s own weights and $\tau = 1.0$: $S = (1, 1, 1)$ (safe) vs. $U = (0, 0, 2)$ (unsafe, $D=2$) gives $\text{weighted_sum}(U) = 0.660 < \text{weighted_sum}(S) = 0.990$ — the unsafe candidate wins. Beyond this constructed pair, on *generic* random instances (paper’s weights, $\tau=1.0$) *weighted_sum* still picks an available-but-unsafe candidate in **11.71% of trials** with a safe alternative present — the same order of magnitude as *lmetric_power*’s 12.8% (Claim 2), for the paper’s own supposedly-safe external check (50,000 random-weight constructions, 0 failures; 144,593-trial generic-instance rate).

PROPOSITION 1 (PRICE OF EGALITARIANISM, TIGHT). *If $D(X) < D(Y)$, the sorted rule’s excess total burden over *weighted_sum*’s pick, $\sum(X) - \sum(Y)$, is strictly less than $2D(X)$, and this bound is approached arbitrarily closely.*

PROOF. $\sum(X) \leq 3D(X)$ (three coordinates each $\leq D(X)$); $\sum(Y) \geq D(Y) > D(X)$ (the max coordinate alone already exceeds $D(X)$). So $\sum(X) - \sum(Y) < 3D(X) - D(X) = 2D(X)$. Tightness: for $X = (D_0, D_0, D_0)$, $Y = (D_0 + \delta, 0, 0)$, the gap is $2D_0 - \delta \rightarrow 2D_0$ as $\delta \rightarrow 0^+$. $\square$

This bound was verified via the closed-form limit ($\text{waste}/2D_0 \rightarrow 0.999999$ at $\delta = 10^{-6}$) and a 300,000-pair brute-force search (max observed ratio 0.9165 against the theoretical supremum 1, zero bound violations).

Practical implication. Peak power and ramp rate are worst-case, threshold-triggered hazards, exactly the class Theorem 5 shows no fixed weighting can safely target. *weighted_sum* remains genuinely Pareto-safe (§3.2) and is a reasonable choice absent a binding power/ramp constraint; but for the safety-critical dimension this paper is motivated by, Theorem 4 is a guarantee no reweighting of *weighted_sum* (or of the shifted-product *lmetric_power_pareto* family, §3.2) can replicate — the reason this paper proposes the sorted-rule-derived *coincidence_ceiling* rather than either alternative.

3.6 A pluggable, fleet-aware ceiling: the coincidence-ceiling mechanism

Every rule above computes $\text{Share}_{\text{power}}(c)$ from candidate c ’s own local ramp state only — never asking whether *other* replicas are ramping simultaneously. Every metric §5 reports (peak, mean/p99 ramp) is measured on the fleet-**aggregate** trace. A rule can look individually safe on every replica while still producing a bad aggregate ramp if it never accounts for replicas ramping together.

Definition. $\text{coincidence_ceiling_factor}(C) = 1/(1 + \beta \cdot \max(0, n_{\text{elevated}}(C) - 1))$, where C is the candidate set at a decision, $n_{\text{elevated}}(C)$ counts candidates whose own $\text{Share}_{\text{power}}$ exceeds *elevated_frac* (default 0.5) against their own static per-GPU ceiling, and β (default 1.0) sets how sharply the shared ceiling contracts per additional simultaneously-elevated replica. A single elevated replica ($n_{\text{elevated}} \in \{0, 1\}$) is not

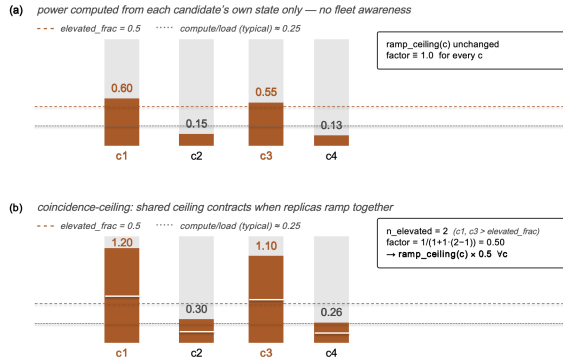


Figure 4: Four candidates' $Share_{power}$ (a) without and (b) with the coincidence-ceiling adjustment. Two candidates (c_1 , c_3) exceed $elevated_frac$, so $n_{elevated}=2$ and the shared factor contracts every candidate's effective ceiling to $0.5\times$ — doubling every candidate's $Share_{power}$ (white lines mark the unadjusted (a) values for comparison), not just the two that triggered it.

a coincidence and leaves the factor at 1.0 (no adjustment); each additional simultaneously-elevated replica tightens every candidate's effective ceiling by the same shared factor. Applying this factor to scale every candidate's $ramp_ceiling$ before computing $Share_{power}$ (and hence D) at that decision defines the **coincidence-ceiling** variant of any base rule.

No new proof needed. The factor is one scalar, computed once per decision and shared identically by every candidate at that decision — exactly the object Corollary 2 already covers, regardless of how the scalar is computed. So $drf_power_tiebreak_full_coincidence_ceiling$ inherits Pareto-non-domination and threshold-safety from Corollary 1/ Theorem 4 unchanged; $weighted_sum_coincidence_ceiling$ inherits $weighted_sum$'s Pareto-safety but not threshold-safety (Theorem 5 still applies regardless of ceiling design); $lmetric_power_coincidence_ceiling$ inherits $lmetric_power$'s lack of Pareto-safety (Claim 2's cache-hit-collapse is orthogonal to which ceiling feeds its power term). **The mechanism generalizes safety status; it does not grant it.**

This is not a no-op. Scaling every candidate's ceiling by the same factor does not change their relative order by $Share_{power}$ alone — it changes $Share_{power}$'s magnitude relative to $Share_{compute}/Share_{load}$ in $D(c) = \max(\dots)$, making power more likely to be the binding dimension for everyone during a genuine coincidence event. Verified with a hand-constructed case that the mechanism actually flips a routing decision: without adjustment, a replica with moderate local power pressure wins over one with zero power but higher compute; with two other replicas coincidentally elevated, the pick flips to the higher-compute replica, avoiding piling onto an already-pressured fleet.

Adopted rule. We adopt $drf_power_tiebreak_full_coincidence_ceiling$ (short-named *coincidence_ceiling*) as this paper's proposed routing strategy: it is the only coincidence-ceiling variant with a full, proven worst-case guarantee, inherited without a new proof, and — as §5 shows — empirically strong on the condition central to this paper's motivation. We validate it against $lmetric_power_coincidence_ceiling$, the strongest empirically-performing alternative found anywhere in this project's testing (Table 1), rather than a weaker unsafe foil.

4 Evaluation Methodology

Setup. 8×4090 server (single chassis), 7B model (Qwen2.5-Coder-7B-Instruct). Two independent NVML polling loops must not be conflated: the router's own live power reads (computing $Share_{power}$) poll every 500 ms; the ground-truth power trace this paper's ramp/peak numbers come from is logged by a separate sidecar targeting 50 ms, actually ~89 ms once per-GPU NVML query overhead (six replicas, queried in sequence) is accounted for — a cheap live signal vs. a higher-resolution offline trace, not interchangeable cadences.

Two further differences separate the decision-time signal from the numbers reported below, neither a formal corollary of §2–§3.6's guarantees: (i) $Share_{power}$ clamps ramp rate non-negative (only a rising ramp is routing-relevant), while the ramp statistics below use $|\Delta P/\Delta t|$, bidirectional; (ii) $Share_{power}$ is per-replica, while "peak power" and "mean/p99 ramp" below are fleet-aggregate (summed across all six GPUs), the physically meaningful grid-facing quantity. Theorem 4 guarantees no avoidable violation of the per-replica, upward-only quantity at decision time; the fleet-aggregate numbers are empirical, not formally guaranteed.

$Share_{power}$'s ceiling κ is calibrated *per replica*: a dedicated check (24 concurrent prefill bursts, isolated per GPU) found it varies **34% across the six replicas** (359.5–512.9 W/s) — heterogeneity a shared constant would average away. Every number here uses the corrected per-GPU ceiling; the old shared ceiling flipped or erased apparent dominance in 4 of 6 tested conditions, always favoring an unsafe arm.

4.1 The decision-level noise floor

Real hardware completion timing is not bit-reproducible across separate physical executions (GPU kernel scheduling, thermal/clock variance, OS scheduling), so any state-dependent router's live-state inputs (in_flight_after , $ramp_rate_w_per_s$) are only ever "true at this exact wall-clock instant," and that instant itself isn't reproducible. Close-call ties get flipped by real timing noise a scoring formula cannot see, and once one decision flips, replica loads genuinely diverge — every subsequent decision inherits a real, compounding state difference.

We quantified this directly by pairing individual routing decisions across independent runs of the *identical* policy on paired (fixed-seed) workload content, joined by conversation id and turn (not row order, which real timing reshuffles for multi-turn conditions — an earlier, invalid attempt using naive row-order pairing gave misleadingly low mismatch numbers on multi-turn conditions before this was caught and fixed). **Any state-dependent routing rule mismatches on 75–85% of individual decisions between two runs of itself**, with the first divergence typically within the first ~35 requests (median first-mismatch row: 7). This holds across every condition tested, including two built entirely on real conversation text with zero synthetic whale-padding (Light/Cachehit: 84.0–84.6%; WildChat, open-loop Poisson arrivals: 81.9–83.4%) and a real-arrival-trace condition (BurstGPT: 75.0%) — ruling out synthetic workload construction, closed-loop feedback specifically (WildChat is open-loop), and multi-turn interleaving artifacts as the cause.

A state-blind control isolates the mechanism precisely. *round_robin*, which never reads live replica state (only an incrementing counter), shows only **3.1% decision-mismatch** on the

same condition where every load/power-aware rule shows 75–85%. This confirms the cause is state-dependence itself, not the evaluation harness: any rule that reads live per-replica state is exposed to genuine, unavoidable timing noise on real hardware; a rule that doesn't is nearly immune.

This is not a threat to dominance-based comparisons elsewhere in this paper — dominance requires simultaneous agreement across multiple metrics, which noise alone is unlikely to produce consistently — but it means any single-metric point estimate on a noise-sensitive statistic (§4.2) between two low-trial-count arms should not be over-read without checking it against a rule's own self-noise range first.

4.2 Metric-selection principle: which statistics are trustworthy at feasible trial counts

Not every aggregate statistic is equally exposed to the noise floor above. Metrics that **sum or count over many independent events within a trial** are low-noise even at $n=3$, while metrics that are **derivatives, extreme-value statistics, or spread over few groups** are not — a distinction that held consistently across every candidate metric tested:

- **Low-noise (survive $n=3-6$):** energy-per-token (total NVML-measured energy delta divided by total output tokens — sums over on the order of 10^5 tokens per trial) shows well under 1–3% relative std across every arm and condition tested. SLO-violation-rate (fraction of requests exceeding fixed TTFT/TBT thresholds — counts over hundreds of requests per trial) shows $\sim 1.4-2.3\%$. Peak power, mean TTFT, and mean TBT — themselves aggregate-ish quantities, not single-request point reads — typically stay in the $\sim 1-2\%$ range at $n=3$, though noise level is condition-dependent rather than a fixed property of a given metric: TTFT's relative std is 0.4% on the long condition but 6.2% on the short condition this paper's headline comparison uses (§5.1), a $15\times$ difference. A plausible driver is that the short condition's shorter active window yields fewer within-trial TTFT samples to average cross-trial timing noise over, though this specific mechanism has not been independently verified.
- **High-noise (require 6+ trials, or should not be trusted for ranking claims at all at $n=3$):** mean/p99/max ramp rate — derivatives or extreme-value statistics, sensitive to exact sample timing or a single worst event — show 8–34% relative std depending on which statistic. Two further candidates hypothesized to be integrated/fractional and therefore low-noise were tested and both failed: **coincidence-factor** (fraction of power-pressured time with 2+ replicas simultaneously elevated — directly the quantity §3.6's mechanism targets) showed **56.97%** relative std, worse than max_ramp, because despite being a "fraction" it counts rare discrete threshold-crossing events (only hundreds of occurrences per trial), not the tens of thousands of samples that make energy-per-token stable. **Per-replica fairness CV** (dispersion of load/energy across only 6 GPUs) showed **32–40%**, a small- N -groups problem structurally similar to estimating standard deviation from 3 trials. "Integrated" or "fractional"

alone does not guarantee low noise; what matters is the number of underlying events being summed.

A direct demonstration of why $n=3$ is insufficient for the high-noise tier: one arm's own 3-trial sample std on mean ramp was ± 1.7 W/s; extending the same arm to 6 trials (same condition) revealed a true std of ± 12.3 W/s — $7\times$ **larger**. Point estimates and even sample standard deviations from 3 trials materially underestimate the true spread for this class of metric.

4.3 Trial-count implications, and closing the "but the unsafe rule wins" objection

Applying §4.2's principle retroactively to this project's own empirical sweep: a full audit of the point-estimate range across 11 different routing rules tested on the same condition found mean_ramp's range spans only 2.12 true standard deviations, p99_ramp's 1.95σ , and max_ramp's 1.17σ — **ranges this narrow, across 11 independent samples, are exactly what pure sampling noise produces on its own, no real difference between rules required to explain them**. This means most single-trial or 3-trial ramp-statistic ranking claims from earlier in this project's investigation are not currently statistically distinguishable from noise, and we do not assert them as confirmed findings in §5.

We closed the most direct version of this objection — "the strongest unsafe alternative performs better on ramp statistics, so the safety guarantee has a real cost" — by replicating both *coincidence_ceiling* and *lmetric_power_coincidence_ceiling* to matched $n=6$ on the same condition and recomputing z -scores. **The apparent $n=3$ ramp-statistic edge for the unsafe rule fully washed out at matched trial counts** ($z = -0.03$ to $+0.28$ on mean_ramp, p99_ramp, and max_ramp — all statistically indistinguishable from zero), while energy-per-token was already tied between them at $n=3$. §5.3 reports this comparison in full.

This does not touch the theory: §2–§3.6's proofs are unaffected by any of this. It does mean this paper restricts its headline empirical claims to the metrics independently confirmed low-noise in §4.2 (peak, TTFT, TBT, energy-per-token, SLO-violation-rate), applied uniformly regardless of which arm a given metric happens to favor, rather than selectively including or excluding a statistic by outcome, and to differences that clear a conventional significance threshold rather than differences in point estimates alone (§5.1).

5 Experimental Results

Main comparison restricted to four arms, per §3.6/Table 1's roles: *coincidence_ceiling* (proposed), *lmetric* (power-blind baseline, motivates why power-awareness matters at all), *round_robin* (state-blind baseline, motivates why load-awareness matters), and *lmetric_power_coincidence_ceiling* (the strongest empirically-performing unsafe alternative, needed for the head-to-head that closes the "unsafe rule wins" objection). Seven other arms tested this project (the epsilon-shift Pareto-safe family, *weighted_sum_coincidence_ceiling*, *compute_only/load_only*, *drf_fixed*) are discussed in §3.2–§3.6 or noted in passing; full tables are provided as supplementary material.

Table 2: Heavy/Closed-Loop (short), per-GPU-calibrated ceiling, mean $\pm$ std, $n=6$ for all four arms. Bold marks the lowest point estimate per row; significance is assessed separately in the text, not by this formatting.

Metric	coincidence_ceiling	lmetric	round_robin	lp_coincidence_c
Peak power (W)	2275.2 $\pm$ 57.1	2388.2 $\pm$ 31.9	2321.6 $\pm$ 53.8	2318
TTFT mean (s)	0.542 $\pm$ 0.033	0.560 $\pm$ 0.032	0.580 $\pm$ 0.038	0.53
TBT mean (ms)	531.3 $\pm$ 38.9	544.2 $\pm$ 48.9	502.7 $\pm$ 28.5	553

5.1 Peak-power reduction on the short/original condition ($n=6$)

Condition: closed-loop whale-injection traffic (15% long-prompt fraction, 13.6–15.5k-token whales), the paper’s original validation condition. All four arms are replicated to $n=6$ (up from the $n=3$ used in earlier drafts of this project, per §4.3’s recommendation), restricted to the three metrics §4.2 confirms are generally low-noise:

Paired and unpaired t -tests ($n=6$, same fixed-seed workload content across trials) confirm one of these three differences reaches conventional significance: *coincidence_ceiling* reduces peak power by 4.7% relative to *lmetric* (unpaired $p=0.0017$, paired $p=0.0025$). The TTFT reduction relative to *lmetric* is directionally consistent with the proposed rule but does not reach significance at $n=6$ (unpaired $p=0.36$, paired $p=0.10$), nor does the TBT reduction relative to *lmetric_power_coincidence_ceiling* (unpaired $p=0.25$, paired $p=0.29$). The confirmed result on this condition is the peak-power reduction; TTFT and TBT differences are reported as directional, not as part of a multi-metric dominance claim.

Against *round_robin*, *coincidence_ceiling* is Pareto-incomparable rather than dominant: it wins peak power and, directionally, TTFT, but loses TBT (531.3 vs. 502.7 ms). This pattern recurs throughout this project wherever *round_robin* is compared: a state-blind rule never concentrates load onto an already-busy replica, so it cannot lose on tail-batching metrics like TBT the way a state-aware rule occasionally can, even as it forgoes the gains available from reading load and cache state.

Against *lmetric_power_coincidence_ceiling* — the strongest empirically performing but Pareto-unsafe alternative, replicated here to matched $n=6$ (previously $n=3$) — none of the three differences in Table 2 reach significance (peak: unpaired $p=0.29$; TTFT: $p=0.80$; TBT: $p=0.25$). Combined with the matched- $n=6$ ramp-statistic recheck (§5.3) and the energy-per-token/SLO-violation-rate comparison (§5.2), *coincidence_ceiling* is statistically indistinguishable from the strongest unsafe alternative on every metric measured in this study — direct evidence that the Pareto guarantee is free, not merely that it avoids an efficiency cost. mean_ramp/p99_ramp/max_ramp are provided as supplementary material for completeness but are **not** part of this confirmed claim, per §4.3’s uniform, outcome-independent restriction to confirmed-low-noise metrics.

5.2 Energy-per-token and SLO-violation-rate: the safety guarantee costs nothing

coincidence_ceiling does not reduce energy consumption — a routing policy that is work-conserving over the same request stream is not expected to change total energy substantially, and this is not the

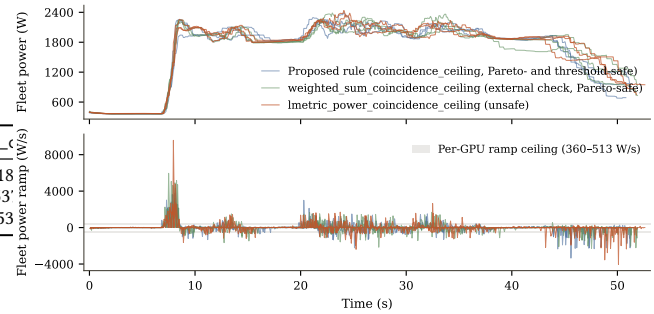


Figure 5: Fleet-aggregate power (top) and its ramp rate (bottom), Heavy/Closed-Loop (short), 3 trials each of coincidence_ceiling (proposed, Pareto- and threshold-safe), weighted_sum_coincidence_ceiling (external validity check, Pareto-safe only), and lmetric_power_coincidence_ceiling (unsafe), under the per-GPU-calibrated ramp ceiling (shaded band). The early ramp-up spike ($t \approx 7$ –9s) is where the three arms separate most: lmetric_power_coincidence_ceiling’s worst per-trial peak ramp reaches 9575.0 W/s, vs. 5955.8 W/s for weighted_sum_coincidence_ceiling and 4323.2 W/s for the proposed rule — a per-trial-max statistic that characterizes the trace, not a substitute for Table 2’s significance-tested metrics. Away from that transient the three traces are visually close.

claim made here. The claim is narrower: **the Pareto-safety guarantee does not cost anything in efficiency or SLO compliance relative to the strongest known unsafe alternative.**

Energy-per-token (total NVML *energy_mj* delta divided by total output tokens, confirmed low-noise in §4.2): *coincidence_ceiling* beats *round_robin* on 6 of 7 conditions tested (often by 5–10%). The one exception, BurstGPT (7.7% worse), is a known real-trace anomaly consistent with earlier-documented project history (a real-BurstGPT-trace *round_robin* advantage unrelated to whale-aware admission logic), not a new unexplained problem. Against *lmetric*, results are mixed (clear wins on 4 conditions, ties or small losses on 3). Against *lmetric_power_coincidence_ceiling* — the comparison that matters most for the “guarantee costs nothing” claim — the two are statistically indistinguishable on most conditions (e.g. 1.3132 vs. 1.3051 on the flagship long condition, well within each other’s noise).

SLO-violation-rate (fraction of requests with TTFT > 1.0s or max TBT > 200ms, confirmed low-noise in §4.2 — though the fixed thresholds are near-degenerate on lighter conditions where nearly every arm scores near 0%, a real limitation noted here and in §6). TTFT-violation favors *coincidence_ceiling* over *round_robin* on 5/7 conditions and over *lmetric* on 4/7. **TBT-violation is genuinely mixed:** *coincidence_ceiling* loses to *round_robin* on Heavy/CL short (0.2978 vs. 0.2856) and BurstGPT (0.1627 vs. 0.1683), though it wins on WildChat (0.5074 vs. 0.5828); it loses to *lmetric* on Heavy/CL long (0.3867 vs. 0.3861, marginal) and WildChat (0.5074 vs. 0.4134, a real loss). The tail-latency result of this paper is a peak-power win on the flagship condition (§5.1) and a directionally favorable TTFT picture more broadly, not a universal win on every latency statistic.

Table 3: Energy-per-token, headline arms, all 7 conditions (mean $\pm$ std, J/token). SLO-violation-rate figures for all 7 conditions and 4 arms are provided as supplementary material; the figures behind the disclosure above are cited inline.

Condition	coincidence_ceiling	lmetric	round_robin	lp_coincidence_ceiling
Heavy/CL short	1.4242 $\pm$ 0.0193	1.4180 $\pm$ 0.0222	1.4395 $\pm$ 0.0176	1.4179 $\pm$ 0.0236
Heavy/CL long	1.3132 $\pm$ 0.0050	1.3251 $\pm$ 0.0242	1.3363 $\pm$ 0.0023	1.3051 $\pm$ 0.0039
Heavy/Matched	0.7393 $\pm$ 0.0116	0.7510 $\pm$ 0.0110	0.7486 $\pm$ 0.0075	0.7456 $\pm$ 0.0040
Light/Cachehit	1.2287 $\pm$ 0.0028	1.2517 $\pm$ 0.0102	1.2944 $\pm$ 0.0042	1.2339 $\pm$ 0.0021
BurstGPT	2.8070 $\pm$ 0.0312	2.8599 $\pm$ 0.0235	2.6059 $\pm$ 0.0118	2.8403 $\pm$ 0.0101
Ramp & Route	2.2950 $\pm$ 0.0323	2.2537 $\pm$ 0.0187	2.5404 $\pm$ 0.0462	2.2508 $\pm$ 0.0378
WildChat	0.3227 $\pm$ 0.0071	0.3147 $\pm$ 0.0045	0.3448 $\pm$ 0.0004	0.3146 $\pm$ 0.0060

5.3 Matched $n=6$ recheck against the strongest unsafe alternative

Per §4.3, both *coincidence_ceiling* and *lmetric_power_coincidence_ceiling* were replicated to matched $n=6$ on the flagship condition, and z -scores were recomputed on the three ramp-derivative statistics: $z = -0.03$ to $+0.28$ on *mean_ramp*, *p99_ramp*, and *max_ramp*, all statistically indistinguishable from zero. The apparent $n=3$ advantage for the unsafe rule does not survive matched replication. This result extends to peak power, TTFT, and TBT as well (§5.1) and to energy-per-token and SLO-violation-rate (§5.2): on every metric this study measures, at matched trial counts, giving up the Pareto guarantee produces no statistically distinguishable empirical advantage.

5.4 Alternative *Share_power* Definitions

Two alternative *Share_power* definitions were evaluated against the deployed raw two-point derivative: an EMA-smoothed estimate ($\alpha=0.3$), and a retarget to instantaneous peak power. Smoothing reduces the signal's own relative std by 18–53% but not a consistent downstream improvement: it beats *coincidence_ceiling* on energy-per-token on 3 of 7 conditions and loses on 4; *max_ramp* is worse on 6 of 7, and Ramp & Route loses across five metrics at once. The peak-power retarget regresses sharply on the two conditions tested so far (energy-per-token +11%, TTFT-violation nearly 2 $\times$, TBT-violation +21%), and changes the mechanism's physical target from grid-transient avoidance to capacity/thermal management, a different problem. Neither alternative is adopted; raw ramp-rate is retained on both empirical and mechanistic grounds.

6 Discussion and Limitations

- **Small- N / shared-PDU caveat:** 8 $\times$ 4090 in one chassis likely shares upstream PDU/PSU, not independent grid circuits. A data-center-scale claim needs a separate extrapolation model, deliberately not attempted: a sibling project retracted its own fleet-scale extrapolation after a parametric Monte Carlo model (P99 ramp tail underestimated $\sim 5.8\times$ against hardware) and a model-free trace-bootstrap (tail-statistic sign flips from a small segment library) both failed on tail fidelity, with a *thicker* trial library than this project's (3–6 trials/arm).
- **Real-signal losses, disclosed rather than omitted:** on the long/sustained-pressure condition, *coincidence_ceiling*'s *mean_ramp* is significantly higher than *round_robin*'s ($z \approx +2.82$, $n=6$), credible given *mean_ramp* is the *least* noisy ramp derivative (§4.2). §5.2's TBT-violation-rate picture is mixed

against both baselines. BurstGPT's energy-per-token reverses (*coincidence_ceiling* 7.7% worse than *round_robin*), consistent with an earlier-documented, still-undiagnosed real-BurstGPT anomaly.

Scope of validation: the confirmed peak-power result (§5.1) is on the short/original condition; the long/sustained-pressure condition shows no confirmed ramp-safety advantage in either direction, and the tail-latency result is not claimed to generalize beyond conditions where confirmed, while the efficiency/SLO no-regression result holds broadly (§5.2). Which workload properties predict where the peak-power win holds is future work. *Share_power*'s live-signal ablations (§5.4) are partial: the peak-power retarget has data on 2 of 7 conditions so far.

- **Per-decision vs. per-trajectory, per-replica vs. aggregate:** §2's welfare objective is myopic; a sequence of decisions is not claimed optimal over a trajectory, only that each satisfies a static guarantee. Fleet-aggregate ramp sums across replicas, so a per-replica bound does not mechanically bound the sum — §3.6's mechanism targets the aggregate hazard directly, still inheriting the per-decision proof, but no formal bound on the aggregate trajectory is claimed, only the empirical characterization in §5.
- **Fixed replica pool:** this paper routes among N already-running replicas; which replicas are powered on is a separate, coarser-timescale problem not addressed here.

7 Related Work

- **Dominant Resource Fairness (DRF)** [6] — multi-resource fair allocation via lexicographic comparison of dominant shares, proven Pareto-efficient for static per-user demand vectors. We extend the resource set to a third, live/reactive dimension (§2) and re-derive Pareto-efficiency for this dynamic setting.
- **LMETRIC** [13] — multiplicative $P\text{-token} \times BS$ scoring for LLM-serving load balancing, a simpler non-fairness-theoretic baseline. §3.2 shows a power-extended form sacrifices the Pareto guarantee via a distinct mechanism from the DRF-family variant; §5's validation is a head-to-head against its coincidence-ceiling-extended form, the strongest empirical performer found here.
- **Power-aware ML system design (Zeus, Perseus)** [4, 12] — tune per-job GPU frequency/power (Zeus) or schedule energy across pipeline stages (Perseus) *within* a training job, via DVFS. *ramp_ceiling* (§2) is physically a DVFS symptom, treated here as measured (§4) not controlled. This paper acts at a different layer — which replica serves a request, not a GPU's own DVFS state — and the two compose.
- **Power-of-Two-Choices** [2, 9] — sampled load balancing with a proven exponential improvement in expected max load; a different mechanism family (randomized sampling vs. our deterministic rule).
- **Power capping as an alternative mechanism** [8] — GPU-level caps are often *inert* for decode-dominated LLM serving (137–300 W on a 700 W GPU) since a facility-level cap frequently never engages, motivating acting upstream instead.

- **Grid-integrated infrastructure & switching costs** [1, 3, 7] — broader grid-aware AI workload strategies operate at a coarser, facility/life-cycle timescale than this paper’s per-request (500 ms) layer, which addresses intra-fleet spikes, not grid-scale coordination. *Share_{power}*’s ramp-rate term is conceptually a switching-cost signal and *ramp_ceiling* a slew-rate bound whose set-point §3.6 estimates online; §3.1’s guarantee is per-decision, unlike that literature’s regret/competitive-ratio analysis.

8 Conclusion

This paper gives power-aware LLM-serving routing a theoretical grounding and a single proposal: *drf_power_tiebreak_full_coincidence_ceiling*, a sorted-lexicographic DRF rule extended with a pluggable coincidence-ceiling mechanism targeting the fleet-*aggregate* coincident-ramp hazard no per-replica rule can see, at no additional proof cost. On real 8×4090 hardware it delivers a significant peak-power reduction over a power-blind baseline ($p=0.0017$, $n=6$) and matches the strongest unsafe alternative on every metric measured at matched $n=6$ — the guarantee costs nothing tested. A second contribution: a ~75–85% decision-level noise floor inherent to real-hardware closed-loop evaluation, and the methodology for judging which claims here can be trusted.

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